

AI-Driven QoS Prediction in SD-WAN Networks



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DECLARATION I, Radhika(Rishiv) Shitlani, hereby declare that this thesis, titled “AI-Driven QoS Prediction in SD-WAN Networks”, and the work presented in it are entirely my own except where explicitly stated otherwise in the text, and that this work has not been previously submitted, in part or whole, to any university or institution for any degree, diploma, or other qualification.

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Abstract

Software-Defined Wide Area Networks (SD-WAN) are used to connect branch offices, cloud services, and data centres over different types of links. A key challenge in SD-WAN is Quality of Service (QoS). Important traffic such as voice or video needs low delay, low jitter, and low packet loss, while background traffic can normally tolerate worse performance. Many SD-WAN policies are still static or reactive. They act after a problem is already visible.

This thesis investigates whether machine learning can predict QoS behaviour in SD-WAN-like environments and support proactive QoS decisions. The work uses two public-data-based experiments. The first experiment uses the 5G Campus Network QoS Dataset for Open-Source gNB Implementations [7] (hereafter, the 5G dataset) to predict achieved throughput from radio and traffic configuration features. This acts as a link-capacity prediction baseline. The second experiment uses a locally generated BNN-UPC/BNNNetSimulator [3] dataset to study per-class QoS delay for Gold, Silver, and Bronze traffic under different scheduling policies.

For the 5G dataset, Random Forest achieved the strongest throughput result among the tested models, with a cross-validated R^2 of approximately 0.94. XGBoost is close behind at 0.93. The dataset is useful for link capacity prediction, but it does not contain QoS class variation.

For the BNN-UPC dataset, XGBoost was compared with two neural models: a PyTorch feedforward MLP and an FT-Transformer. All three models achieved similar performance on the log-transformed average delay target, with cross-validated R^2 values around 0.766–0.769, and the per-class results converged to the same Bronze ceiling.

The thesis also addresses a key evaluation issue: a single global metric can hide poor performance for one QoS class. Therefore, the BNN-UPC model is evaluated by QoS class, scheduling scenario, and scheduling policy. The results show that Gold and Silver delay are predicted much more accurately than Bronze delay, with per-class R^2 values of 0.914, 0.908, and 0.704 respectively on a lightly loaded dataset.

Beyond average delay, the evaluation is extended to two additional QoS targets: 90th-percentile delay (delay_p90) and jitter. delay_p90 shows moderate predictability for Gold and Silver ($R^2 \approx 0.64$ – 0.66), while jitter is found to be largely unpredictable from static network configuration features ($R^2 < 0.12$). This is a meaningful finding: jitter is driven by dynamic traffic burstiness that queue weights and scheduling policy alone cannot capture.

The Bronze SLA violation threshold is tuned from 100 ms to 60 ms based on a threshold sweep, improving Bronze Recall from 0.368 to 0.568 and F1 from 0.489 to 0.592. Packet loss evaluation is restricted to Bronze traffic, which is the only class that experiences meaningful loss in a realistic network (10.7% of Bronze flows in v1 have non-zero loss; Gold and Silver have near-zero loss). A binary classifier (predicting any loss versus no loss) achieves AUC-ROC of 0.916 for Bronze flows, confirming the model can discriminate loss-prone flows

well even though exact loss-rate regression is not feasible from static features. Finally, a Layer 3 allocation recommender is implemented. It tests candidate WFQ bandwidth profiles and recommends the least-violating Gold/Silver/Bronze allocation under SLA thresholds. This connects QoS prediction to the original SD-WAN decision question: how much bandwidth should be allocated to each QoS class?

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Chapter 1

Introduction

In today's digital world, businesses rely heavily on wide area networks (WANs) to connect their offices, data centers, and cloud services. Traditional WANs, which often use expensive and rigid MPLS links, are increasingly being replaced by SD-WAN. SD-WAN offers more flexibility, cost savings, and centralized control. One of the biggest responsibilities of any WAN including SD-WAN is to ensure smooth and reliable network performance. This is referred to as QoS, which includes important metrics like delay (latency), jitter, packet loss, and available bandwidth.

In networks, traffic patterns keep fluctuating since different applications have different needs, and the same network might suddenly be overloaded. For example, video conferencing needs minimal or zero packet loss along with low latency, while file downloads just need higher bandwidth. If this is not taken care of, user experience is impacted and organizations may face penalties for violating SLAs(Service Level Agreements). Therefore, maintaining good QoS is not only critical but challenging.

Currently, most SD-WAN solutions use static policies to configure QoS, either set manually in advance or adjusted only after a problem is detected. Static poli-

cies cannot adapt to rapidly changing traffic conditions, and as a result they often violate application SLAs precisely when demand spikes: by the time a QoS issue is noticed, the user experience is already degraded. This is especially damaging for real-time applications such as voice and video calls, which need immediate action. There is therefore a clear need for a predictive, adaptive approach that can anticipate QoS problems before they occur rather than reacting after the fact.

This is where Artificial Intelligence (AI) and Machine Learning (ML) come in and can help to resolve the problem. These technologies can learn from historical data and detect patterns in the network traffic. If trained correctly, ML models can predict QoS problems before they happen. For example, if the system learns that high video traffic around 2 PM usually leads to congestion, it can automatically adjust bandwidth or reroute traffic ahead of time. This kind of prediction allows network operators to prevent problems instead of just reacting to them.

Scope and Context

This thesis explores how AI and ML techniques can be used to predict QoS in SD-WAN-like networks. The project began with a broad focus on QoS prediction, but the implemented work now has three connected layers. Layer 1 predicts link capacity using the 5G dataset [7]. Layer 2 predicts per-class delay using a BNN-UPC/BNNNetSimulator [3] dataset, using XGBoost as the strongest explainable tree-based baseline and comparing it against two neural models: a feedforward MLP and an FT-Transformer. Layer 3 uses the delay model to recommend a WFQ bandwidth allocation for Gold, Silver, and Bronze traffic. Figure 1.1 shows this layered architecture and how the layers connect.

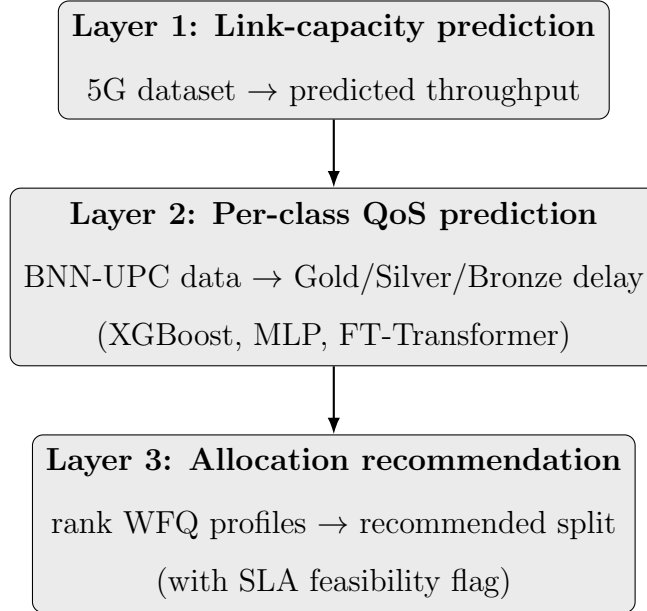


Figure 1.1: The three-layer architecture of the thesis. Each layer produces an output consumed by the next: link-capacity prediction sets the context, per-class delay prediction is the core modelling task, and the allocation recommender turns predictions into a bandwidth-allocation decision.

Problem Statement

SD-WAN makes it easier to manage WANs, but maintaining QoS is still a challenge, especially during unexpected traffic surges or link failures. Most QoS-related decisions today are made either too late or based on static rules. Predicting QoS metrics in advance using AI could solve this problem but it's not clear which methods work best, how accurate they are, or how easy they are to implement in real networks.

Research Questions

This thesis aims to answer the following questions:

-
- **RQ1:** Can supervised machine learning accurately predict per-class QoS metrics (average delay, tail delay, jitter, and packet loss) in an SD-WAN-like environment?
 - **RQ2:** Which model family, tree-based machine learning (XGBoost) or deep learning (a feedforward MLP and an FT-Transformer), offers the best balance of predictive accuracy, interpretability, and training/inference cost for this task?
 - **RQ3:** Can the predicted per-class QoS be used to support an automated bandwidth-allocation decision?

These questions map directly onto the three implemented layers: RQ1 and RQ2 are answered by the per-class delay-prediction experiments (Layer 2), and RQ3 by the allocation recommender (Layer 3).

Methodology

To explore these questions, a literature review was conducted using academic databases such as IEEE Xplore, Scopus, and Google Scholar, searching for terms such as “QoS prediction,” “SD-WAN,” “machine learning in networking,” and “AI-based traffic forecasting.” Papers published between 2020 and 2025 were prioritized, and older foundational works were included when relevant.

The implementation then followed an experimental approach. First, public datasets were reviewed and unsuitable proxy datasets were removed from the main modelling path. Second, supervised ML models were trained on the 5G dataset and the BNN-UPC dataset, including classical ML models, a feedforward MLP, and an FT-Transformer. Third, the BNN-UPC results were evaluated using QoS-aware metrics, including per-class error, policy-specific slicing, RMSLE, and SLA violation precision. Finally, a simple allocation recommender was built to

connect prediction to an SD-WAN control decision. The allocation recommender currently uses the XGBoost delay model; extending it to the neural models is left as future work.

The literature review and background material in this thesis are secondary research, summarising work already published by others. The dataset generation, preprocessing, modelling, QoS-aware evaluation, and allocation recommender are the author’s own primary research and constitute the contribution of this work.

Contributions of this Thesis

The main contributions of this thesis are:

1. **A layered AI framework for SD-WAN QoS decision support**, connecting link-capacity prediction (Layer 1), per-class QoS prediction (Layer 2), and bandwidth-allocation recommendation (Layer 3) into a single pipeline.
2. **A reproducible per-class QoS dataset** generated with the official BN-NetSimulator, providing Gold/Silver/Bronze delay, jitter, and loss under different scheduling policies, together with the code to regenerate it.
3. **A QoS-aware evaluation methodology** that goes beyond global MAE/RMSE/ R^2 to per-class error, RMSLE, scheduling-policy and scenario slices, and SLA-violation precision/recall, exposing class-level behaviour that a single global metric hides.
4. **A controlled comparison of tree-based machine learning and deep learning** (XGBoost versus an MLP and an FT-Transformer) under an identical out-of-fold protocol, including fold-level variability and a paired statistical significance test.

-
5. **An AI-assisted WFQ allocation recommender** that uses the delay model as a surrogate to rank candidate bandwidth profiles and, importantly, flags when no profile can satisfy the SLA, turning prediction into an actionable, and honestly bounded, control decision.

Structure of the Document

This document is structured as follows:

- Chapter 1 introduces the research topic, its scope, the problem statement, and the key research questions.
- Chapter 2 provides background information on SD-WAN, QoS metrics, and the role of AI in modern network management.
- Chapter 3 presents the literature review, including an analysis of academic work on QoS prediction in SD-WAN using AI and ML techniques.
- Chapter 4 outlines the methodology, including dataset selection, feature engineering, model design, QoS-aware evaluation, and the allocation recommender.
- Chapter 5 presents the experimental setup and results for the 5G dataset and the BNN-UPC dataset.
- Chapter 6 discusses why the results occurred, including why some QoS classes are harder to predict and why deep learning did not outperform the tree-based model.
- Chapter 7 states the limitations of the work, including data realism, the static-feature ceiling, and offline evaluation.

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- Chapter 8 concludes the thesis by summarising findings, discussing impact, and identifying directions for future work.

Chapter 2

Background

This chapter provides foundational knowledge needed to understand the scope and significance of this thesis. It covers the key concepts of SD-WAN, QoS metrics in networking, and the role of AI in network management. These concepts establish the context for the research discussed in later chapters.

2.1 SD-WAN

SD-WAN is an approach to wide area networking that uses software-based control to manage network traffic over multiple connection types, including MPLS, broadband, LTE, and 5G. Unlike traditional WANs that rely on expensive private circuits, SD-WAN provides centralized orchestration and greater flexibility, enabling cost-effective routing based on application priorities and real-time conditions.

A typical SD-WAN architecture includes:

- **Edge devices:** Deployed at branch locations to route traffic intelligently.
- **Central controller:** Makes policy-based routing decisions and pushes configurations to edge devices.

- **Overlay and underlay networks:** Logical tunnels (overlay) established on top of physical connections (underlay).

SD-WAN improves application performance by enabling dynamic path selection, fast failover, traffic shaping, and visibility across all network segments.

2.2 QoS

QoS in networking refers to the ability to provide different priority levels to different applications, users, or data flows, and to guarantee a certain level of performance for critical traffic. It is crucial for maintaining service-level agreements (SLAs), especially in modern, cloud-centric environments.

Common QoS metrics include:

- **Latency:** Time taken for a packet to travel from source to destination.
- **Jitter:** Variation in packet delay across a network.
- **Packet loss:** Percentage of packets lost during transmission.
- **Bandwidth:** Maximum data transfer rate of a link.

Effective QoS management ensures that delay-sensitive applications like VoIP and video conferencing are not negatively affected by congestion or fluctuating traffic.

2.3 AI in Network Management

AI, especially ML, is increasingly being used in networking to automate and optimize operations. AI techniques can analyze large volumes of network telemetry data to detect anomalies, classify traffic, and predict future network states.

In the context of SD-WAN, AI is being used to:

2.4 Importance of QoS Prediction in SD-WAN

- Predict traffic trends and adjust policies proactively.
- Recommend optimal routing paths based on current and expected conditions.
- Forecast QoS metrics and warn of potential SLA violations.

AI models used for QoS prediction typically include supervised learning algorithms (like decision trees and neural networks) and reinforcement learning (used in adaptive routing and load balancing). These models require training on historical network data to make accurate forecasts.

2.4 Importance of QoS Prediction in SD-WAN

Modern enterprises rely on applications that require consistent and reliable network performance. Static or reactive QoS mechanisms are often too slow to prevent performance issues. QoS prediction enables a more proactive approach. By forecasting issues before they occur, it becomes possible to take preventive actions like re-routing traffic or scaling resources.

This capability is especially important in SD-WAN environments where network conditions vary across different links and access types. Accurate QoS prediction can help optimize resource usage, reduce costs, and improve user experience.

Table 2.1 contrasts the traditional, rule-based approach to SD-WAN QoS management with the AI-assisted approach this thesis investigates. The central shift is from static, reactive control to learned, proactive decision support.

2.4 Importance of QoS Prediction in SD-WAN

Table 2.1: Traditional versus AI-assisted SD-WAN QoS management.

Traditional SD-WAN	AI-assisted SD-WAN
Static configuration	Predictive from data
Reactive (acts after a problem)	Proactive (anticipates a problem)
Manual, expert-tuned policies	Learned from historical behaviour
Fixed queue weights	Adaptive, per-class allocation
One global setting	Class-aware (Gold/Silver/Bronze) decisions

Chapter 3

Literature Review

This chapter presents a detailed review of the existing research related to the use of AI and ML techniques for predicting QoS in SD-WAN. The aim is to understand what work has already been done, what methods and models have been proposed, and where gaps still exist. This will help assess the current state of the field and how it supports or challenges the research questions introduced earlier.

3.1 Search Strategy and Selection Process

To gather the literature for this review, a structured search was carried out using academic databases such as IEEE Xplore, Scopus, and Google Scholar. The following search terms were used:

- “QoS prediction” AND “SD-WAN”
- “AI for QoS” AND “Software-defined networking”
- “Machine learning traffic classification” AND “network performance”

3.2 Overview of Selected Papers

The search focused on papers published between 2018 and 2025 to ensure up-to-date relevance. Articles before 2018 were excluded unless they provided foundational knowledge. From an initial pool of documents, key research papers were selected based on the following criteria:

- Peer-reviewed publications
- Clear focus on QoS, SD-WAN, or software-defined networking
- Use of AI or ML for prediction, classification, or optimization
- Availability of experimental results, models, or architecture designs

Although the topic of QoS prediction using AI in SD-WAN is promising, the literature review revealed a limited number of high-quality research papers that directly focus on this specific intersection. Most existing studies either concentrate on general SDN environments or address QoS management without the use of AI or ML techniques. Nonetheless, this review provides valuable insights and highlights key research gaps that warrant further investigation.

3.2 Overview of Selected Papers

The selected papers cover a range of methods and applications: some focus on predicting traffic load, others on routing or load balancing. Table 3.1 categorizes these papers by their core approach and relevance to the research questions.

3.3 AI and ML Methods for QoS Prediction (RQ1)

Many studies have applied AI techniques to network traffic prediction and management. For instance, Rahman et al. (2018) [8] used Random Forest and deep

3.4 Suitability and Real-World Limitations (RQ2)

learning models to predict how many virtual network functions (VNFs) are needed during a traffic surge. Their models achieved over 95% accuracy and could prevent service degradation by scaling resources in advance.

Similarly, Zheng et al. (2022) [10] used supervised machine-learning techniques to classify traffic flows according to application type and QoS requirements. They then routed this traffic using a QoS-aware routing algorithm that selected paths depending on the required delay tolerance. This combination of classification and path assignment showed how AI can directly improve QoS outcomes.

These studies clearly support RQ1 by showing that supervised learning and deep learning can be applied to anticipate traffic needs and assign optimal resources in advance.

3.4 Suitability and Real-World Limitations (RQ2)

Although the above techniques work well in controlled settings, there are challenges when using them in real networks. For example, the deep reinforcement learning (DRL) model by Dinh et al. (2024) [2] performs well in simulations and maintains safe traffic routing using control barrier functions (CBFs). However, DRL requires substantial training time and often benefits from GPU acceleration, which may not be available in all SD-WAN deployments.

In contrast, Quang et al. (2023) [6] take an analytical rather than a learned approach. Their central contribution is SABE, a bandwidth estimator that infers how much capacity is actually free on each overlay link *without sending probe traffic*. SABE models each link as an M/M/1/K queue and inverts the standard queuing-theory equations for delay and loss: because SD-WAN routers already

3.4 Suitability and Real-World Limitations (RQ2)

report measured delay and loss, these measurements are used to recover the total load on the link, and subtracting the SD-WAN’s own traffic yields the hidden cross-traffic and hence the available bandwidth. This estimate then feeds a two-loop controller: a slower loop optimises routing (path split ratios) and a faster loop tunes per-class QoS policies (queue weights and shaper rates) using a lightweight local-search optimiser instead of a slow exact solver. Traffic is grouped into four classes (Critical, VoIP, Office, Bulk), each with its own delay and loss target.

In their NS-3 simulation, combining routing optimisation, QoS optimisation, and SABE raised SLA satisfaction for critical traffic by roughly 40% over static policies, with the bandwidth estimate doing most of the work: optimisation alone moved critical-class satisfaction from about 55% to 64%, while adding SABE pushed it to about 98%. The local-search optimiser matched the exact solver almost exactly (97.7% versus 97.8%) at much lower cost.

This work is directly relevant to the present thesis, but it is important to note what it is *not*: SABE is a fixed, model-based estimator, not a learned one. Its accuracy rests on the M/M/1/K assumption of Poisson arrivals and exponential service times, which does not describe the bursty, self-similar traffic of real WANs; this is precisely the assumption a data-driven model could relax. SABE therefore serves as a strong non-learned baseline that motivates the machine-learning direction taken here. More generally, this contrast shows that AI models are powerful but their complexity and resource requirements must be weighed against simpler analytical methods before deployment, which directly addresses RQ2.

3.5 Integration into SD-WAN Systems (RQ3)

Several papers also discussed how these models can be integrated into actual SD-WAN systems. For instance, Rahman et al. (2018) [8] simulated an SD-WAN scenario where their ML models helped reduce both QoS violations and cloud leasing costs by scaling VNFs only when necessary.

Quang et al. (2022) [5] presented a practical example of an enterprise moving from traditional MPLS to SD-WAN. Their central controller dynamically adjusted both routing and QoS scheduling based on the current network status. This shows that optimisation-based decision-making can be realistically applied in SD-WAN setups to improve efficiency and reliability, even when the control logic is not a learned predictor.

These findings support RQ3, confirming that automated decision support is not only a research idea but can also be integrated into live systems if the environment supports it.

3.6 Comparative Analysis and Research Gaps

Rather than summarising each paper in isolation, Table 3.2 compares the reviewed works along the dimensions most relevant to this thesis: the data source, the modelling approach, the QoS aspect addressed, and the main limitation relative to the present work.

3.6 Comparative Analysis and Research Gaps

Table 3.2: Comparative analysis of the reviewed literature. Dataset descriptions are characterised from the cited works and should be read at the level of data *type*.

Work	Data	Model	Gap relative to this thesis
Rahman et al. [8]	Simulation	RF + DNN	VNF autoscaling only; no per-class QoS prediction
Zheng et al. [10]	SDN simulation	Supervised ML + QoS routing	Classification + routing; single metric, no class-wise error
Dinh et al. [2]	Simulation	DRL (PPO/DDPG)	Control-focused; heavy compute; no interpretable prediction
Quang et al. [6]	NS-3 sim.	SABE (queuing) + local search	Model-based estimator + optimisation, not a learned predictor; no ML/DL comparison
Quang et al. [5]	SD-WAN sim.	Centralised optimisation	Optimisation, not prediction; no ML model comparison
Chen et al. [1]	Simulation	GNN + MARL	Complex control pipeline; not a reproducible per-class study
This thesis	Reprod. sim.	XGBoost, MLP, FT-Transf.	Per-class QoS-aware evaluation + DL-vs-ML + allocation recommender

3.6 Comparative Analysis and Research Gaps

Reading across the table, a consistent pattern emerges. Most prior works: (i) rely on simulation with no reproducible public dataset; (ii) predict or optimise a single QoS dimension rather than several; (iii) do not treat Gold/Silver/Bronze traffic classes explicitly, so class-level failures stay hidden behind global metrics; (iv) do not compare deep learning against tree-based machine learning on the same task; and (v) stop at prediction or control without connecting the model back to an explicit bandwidth-allocation decision. These five gaps directly motivate the contributions of this thesis, listed in Section 1: per-class QoS-aware evaluation, a reproducible simulator-generated dataset, a tree-based versus deep-learning comparison, and an AI-assisted bandwidth-allocation recommender.

The following specific observations were also noted:

- Most research predicts a single QoS metric (latency or bandwidth) rather than several at once.
- Most results are simulation-based; live SD-WAN deployment results are rare.
- Public SD-WAN QoS datasets are scarce, which limits reproducibility.
- Few works compare AI predictions directly against the decisions a standard SD-WAN controller would make.

Of these four gaps, this thesis directly closes two. It predicts multiple QoS metrics (average delay, tail delay, jitter, and packet loss) rather than a single one, and it releases a reproducible dataset generated with BNNetSimulator together with the code needed to regenerate it. The remaining two gaps are not closed by this thesis: like the works reviewed above, its results are simulation-based rather than measured on a live SD-WAN (Chapter 7 discusses this explicitly), and the Layer 3 recommender in Chapter 5 ranks candidate bandwidth profiles against

each other using the trained model rather than against the decision a standard, rule-based SD-WAN controller would actually make. Filling these gaps could make QoS prediction models more robust, more generalizable, and more useful for industry; the controller-comparison gap is carried forward explicitly as future work (Chapter 8).

3.7 Summary of the Literature Review

This review found that AI and ML models, especially supervised learning, deep learning, and reinforcement learning, are promising tools for predicting QoS trends in SD-WAN. The models help forecast traffic load, classify application needs, and adjust routing or resource allocation dynamically. However, practical deployment still faces challenges due to training complexity, hardware requirements, and a lack of real-world validation.

Although the number of direct studies focused on AI-based QoS prediction in SD-WAN remains limited, the relevance of the topic continues to grow as networks become more software-defined and application-aware. Recent work such as MTEAL [1] shows that state-of-the-art SD-WAN traffic engineering is moving beyond simple machine learning towards graph neural networks, reinforcement learning, and multi-dimensional QoS optimisation. Informed by this review, the present work uses simple models as baselines and additionally evaluates stronger tree-based and deep-learning models (XGBoost, a feedforward MLP, and an FT-Transformer). The scope was deliberately kept realistic for a master's-level project: the comparison focuses on per-class delay prediction on a reproducible dataset rather than on full reinforcement-learning-based control.

3.7 Summary of the Literature Review

Table 3.1: Summary and Categorisation of Selected Literature

Paper	Technique / Method	Main Focus	Related RQ(s)
Rahman et al. [8]	Random Forest, DNN	Predicting VNF requirements to maintain QoS and reduce cost	RQ1, RQ2, RQ3
Zheng et al. [10]	Supervised ML + QoS routing	Classifying application types and assigning QoS-based routes	RQ1, RQ2
Dinh et al. [2]	DRL (PPO, DDPG) with CBF	Safe traffic routing using AI that respects network constraints	RQ1, RQ2
Quang et al. [6]	SABE (Queuing theory-based estimation)	Estimating available bandwidth using passive measurements	RQ2, RQ3
Quang et al. [5]	Centralised Optimization + QoS scheduler	Managing QoS priorities and routing using SD-WAN controller	RQ2, RQ3
Chen et al. [1]	GNN + MARL	SD-WAN traffic engineering with multi-dimensional QoS metrics	RQ1, RQ2, RQ3

Chapter 4

Methodology

This chapter describes the method used in this project. The aim is not to build a commercial SD-WAN product. The aim is to test whether machine learning can predict QoS outcomes and whether those predictions can support a simple SD-WAN bandwidth-allocation decision.

The project uses a layered methodology:

- **Layer 1: Link capacity prediction.** The 5G Campus Network QoS Dataset for Open-Source gNB Implementations [7] (the 5G dataset) is used to predict achieved throughput from radio and traffic configuration features.
- **Layer 2: Per-class delay prediction.** A BNN-UPC/BNNetSimulator [3] dataset is used to predict delay for Gold, Silver, and Bronze traffic classes under different scheduling policies.
- **Layer 3: Allocation recommendation.** The trained delay model is used as a surrogate model to test candidate WFQ weight profiles and recommend a Gold/Silver/Bronze bandwidth allocation.

All data-generation, preprocessing, modelling, and evaluation code, together

with the generated BNN-UPC dataset, is available in the project’s public code repository.¹

Figure 4.1 summarises the end-to-end experimental workflow that produces the results in Chapter 5. The same pipeline is applied to both datasets, with only the feature set and target changing between them.

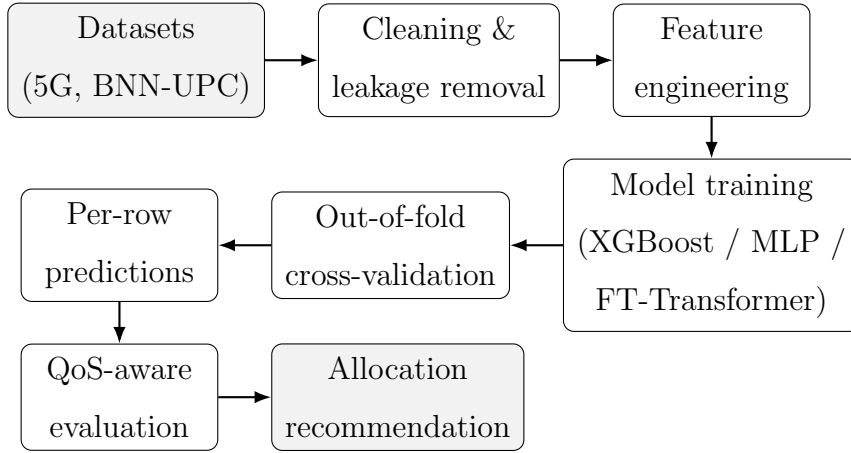


Figure 4.1: End-to-end experimental workflow, from raw datasets to the allocation recommendation. Leakage columns are removed before feature engineering; models are compared under an identical out-of-fold cross-validation protocol; and the resulting per-row predictions feed both the QoS-aware evaluation and the Layer 3 recommender.

4.1 Dataset Selection

A real enterprise SD-WAN dataset was not available. Therefore, public datasets and simulation-based datasets were used.

¹Code and generated data: <https://github.com/rishivshitlani/sdwan-qos-prediction>.

4.1.1 5G Dataset

The 5G dataset used in this thesis is officially titled “5G Campus Network QoS Dataset for Open-Source gNB Implementations” [7] (hosted on Zenodo, record 13754300). It contains real 5G throughput measurements from two testbeds. It includes offered throughput, achieved throughput, jitter, packet loss, radio bandwidth, slot configuration, uplink/downlink direction, and one-way-delay packet measurements.

This dataset is useful for predicting link capacity. However, it has an important limitation for this thesis: all rows belong to the same application type. It does not contain Gold, Silver, and Bronze QoS class variation. Therefore, it cannot answer the per-class QoS allocation question by itself.

4.1.2 BNN-UPC / BNNetSimulator Dataset

The second dataset was generated locally using the official BNNetSimulator [3] Docker image, published by the Barcelona Neural Networking Center (BNN-UPC) at Universitat Politècnica de Catalunya. This was necessary because the official BNN-UPC Challenge download links were not available. BNNetSimulator is the same type of simulator used to generate the BNN-UPC network modelling datasets, so it is suitable for a reproducible simulation-based QoS experiment.

The generated BNN-UPC dataset contains 29,280 per-flow records. Each row represents one source-destination flow. The dataset includes three QoS classes:

- **Gold:** ToS 0, highest priority.
- **Silver:** ToS 1, medium priority.
- **Bronze:** ToS 2, lowest priority.

Four scheduling scenarios were generated:

- Fixed WFQ weights.
- Random WFQ profiles.
- Mixed SP/WFQ/DRR policies with skewed ToS distribution.
- Mixed SP/WFQ/DRR policies with equal ToS distribution.

This dataset is referred to as the **v1 dataset** and is used for the main QoS scheduling and allocation experiments under a light-to-moderate load regime.

4.1.3 Datasets Not Used as Main Inputs

CICIDS2017 was explored earlier but was dropped from the main modelling path. It is an intrusion-detection dataset, not a QoS dataset. Using it would require too much reverse engineering of QoS labels. MAWI was also considered, but it does not contain direct Gold/Silver/Bronze labels and would require substantial extra feature engineering. Therefore, the final implemented work focuses on the 5G dataset and BNN-UPC.

4.2 Feature Engineering

For the 5G dataset, the main target is `actual_throughput_mbps`. The input features include radio configuration, offered throughput, direction, testbed metadata, latency, packet count, and flow duration. Leakage columns such as `recommended_bandwidth_percent`, `jitter_ms`, and `packet_loss_percent` are removed when predicting throughput.

For the BNN-UPC dataset, the main target is `log_avg_delay`. The log transform is used because raw `avg_delay` is heavy-tailed. Bronze traffic can experience much larger delay spikes than Gold and Silver traffic, especially under congestion.

The BNN-UPC feature set used for modelling contains:

- `tos` and `qos_class`
- `offered_bandwidth`
- `time_distribution`
- `routing_hops`
- `n_nodes`
- `max_avg_lambda`
- `link_bandwidth`
- `scheduling_policy`
- `tos_queue_weight`
- `min_tos_weight`

Outcome columns such as `avg_delay`, `jitter`, `packet_loss_rate`, delay percentiles, and `actual_bandwidth` are removed from the input features to avoid leakage.

4.2.1 Feature Importance (SHAP)

To understand which features drive the delay prediction, SHAP (SHapley Additive exPlanations) values were computed for the trained XGBoost model. SHAP attributes each prediction to its input features and, unlike the model's built-in gain importance, also shows the *direction* of each effect. Figure 4.2 shows the SHAP summary for log delay.

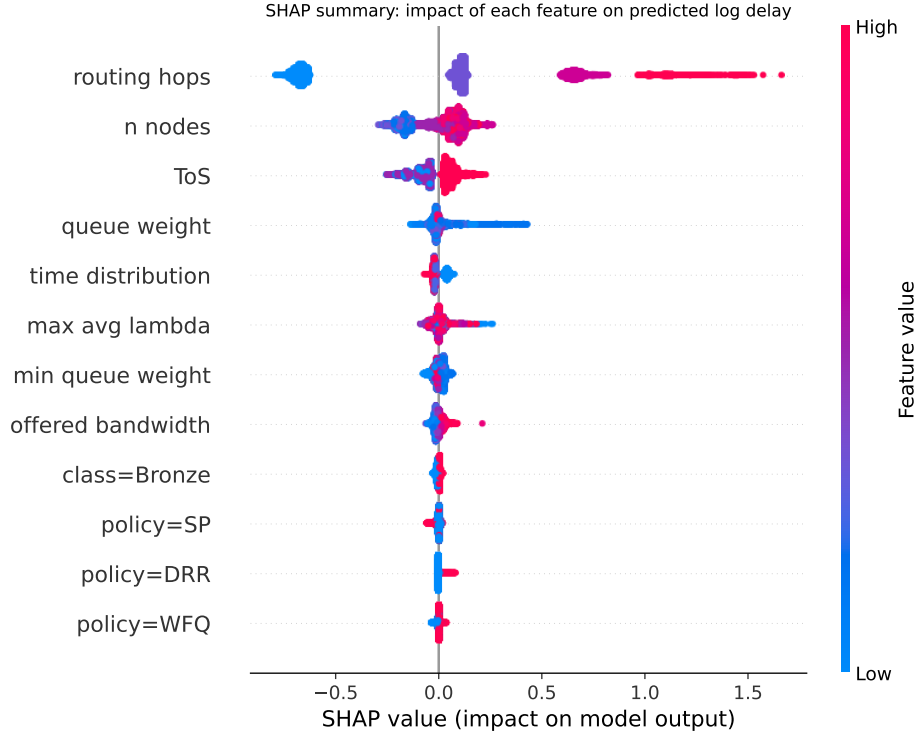


Figure 4.2: SHAP summary for the XGBoost log-delay model. Each point is a flow; position shows how much a feature pushed the prediction up (right, more delay) or down (left, less delay), and colour shows the feature value (red high, blue low).

The dominant driver is **routing hops**: flows with more hops (red) consistently receive a large positive push towards higher delay, while short paths (blue) reduce it. Network size (**n_nodes**) and traffic class (**tos**) follow. This is physically sensible, since delay accumulates at each hop along a path. Queue weight, scheduling policy, and QoS-class indicators have comparatively small impact. The strong dependence on path structure, and the weak dependence on the dynamic-load features available here, is consistent with the later finding that the static feature set captures mean delay well but not the burst-driven variance of Bronze traffic.

4.3 Models

The following models were trained and compared:

- **DummyRegressor:** A naive baseline that predicts the training mean.
- **Linear Regression:** A simple interpretable baseline.
- **Support Vector Regression (SVR):** A kernel-based non-linear model.
- **Random Forest Regressor:** A tree ensemble model.
- **XGBoost Regressor:** A gradient-boosted tree model.
- **PyTorch MLP:** A feedforward neural network with hidden layers of 128, 64, and 32 neurons.
- **FT-Transformer:** A modern attention-based architecture for tabular data (Feature Tokenizer + Transformer), which tokenises each feature and applies self-attention across the feature tokens.

The simple ML models are important because they are easier to explain and are suitable for tabular data. XGBoost is treated as the strongest explainable tree-based baseline. The PyTorch MLP and the FT-Transformer represent the neural-network direction of the project: the MLP tests whether a standard feedforward network can learn the same QoS delay behaviour, while the FT-Transformer tests whether a modern, attention-based deep-learning architecture designed for tabular data can outperform the tree-based baseline.

4.4 Evaluation Methodology

The first evaluations used common regression metrics, defined here for clarity:

- **Mean Absolute Error (MAE):** the average absolute difference between predicted and actual values. It is easy to interpret and relatively insensitive to outliers.
- **Root Mean Squared Error (RMSE):** the square root of the mean squared error. It penalises large errors more heavily than MAE.
- **R^2 score:** the proportion of variance in the target that the model explains. A value of 1.0 is perfect, 0.0 is no better than predicting the mean, and negative values are worse than the mean. Five-fold cross-validated R^2 is also reported to check stability.
- **Inference time:** the time taken to produce a prediction. This matters for SD-WAN, where decisions must be made in near-real time.

These global metrics are necessary but not sufficient for BNN-UPC. Gold, Silver, and Bronze traffic have different priorities and different delay behaviour, and a single global metric can hide poor performance for one class.

Therefore, the BNN-UPC evaluation was extended using the following methods:

- **Per-class MAE:** Errors are reported separately for Gold, Silver, and Bronze flows.
- **Policy-specific slicing:** Results are sliced by scheduling scenario and scheduling policy.
- **RMSLE:** Root Mean Squared Logarithmic Error is reported because delay has a heavy-tailed distribution.
- **SLA violation precision:** Continuous delay predictions are converted into binary SLA violation triggers. Precision measures how trustworthy the trigger is when the model predicts a violation.

This evaluation better matches the SD-WAN use case. It shows not only whether the model is accurate overall, but also whether it is reliable for high-priority traffic.

4.4.1 Multi-Metric QoS Evaluation

Average delay is the primary QoS target, but real SD-WAN environments also care about tail delay and jitter. Therefore, the evaluation is extended to two additional targets:

- **delay_p90:** The 90th percentile of packet delay. It represents the worst-case experience for 10% of packets, which is more relevant than mean delay for real-time applications.
- **jitter:** Variation in packet delay. High jitter degrades voice and video quality even when mean delay is acceptable.

Both targets are heavy-tailed like average delay, so a log transformation is applied before training and predictions are back-transformed for reporting. MAE, RMSE, RMSLE, and R^2 are reported per QoS class for each target. This allows direct comparison of how predictable each metric is from the available network configuration features.

4.5 Layer 3 Allocation Recommender

The final implemented component is a simple allocation recommender. It is model-agnostic: it can use any trained BNN-UPC delay model (XGBoost, the MLP, or the FT-Transformer) to test candidate WFQ profiles such as:

- 60/30/10

4.5 Layer 3 Allocation Recommender

- 50/40/10
- 33/33/34
- 25/65/10
- 80/15/5

Each profile represents a possible Gold/Silver/Bronze bandwidth allocation. For each profile, the recommender rewrites the queue-weight features, predicts class-level delay, checks SLA thresholds, and ranks the profiles. The reported allocation results use the trained XGBoost delay model; running the same workflow with the neural models is left as future work. This directly addresses the original SD-WAN question: how much bandwidth should be allocated to each QoS class?

Chapter 5

Experiments and Results

This chapter presents the experiments completed so far. The results are organised by the three project layers: 5G throughput prediction, BNN-UPC per-class delay prediction, and Layer 3 QoS allocation recommendation.

5.1 Layer 1: 5G Throughput Prediction

5.1.1 Dataset and Setup

The 5G dataset [7] was used to predict achieved throughput. The processed dataset contains 604 rows after uplink and downlink measurements are split into separate records. The target is `actual_throughput_mbps`.

The dataset was split into 80% training and 20% testing using a fixed random seed. Five-fold cross-validation was also used on the training split. Leakage columns such as `recommended_bandwidth_percent`, `jitter_ms`, and `packet_loss_percent` were dropped. Constant metadata columns such as `protocol`, `application_type`, `link_type`, and `sdr` were also removed because they do not vary across the dataset.

5.2 Layer 2: BNN-UPC Delay Prediction

5.1.2 Results

Table 5.1 shows the 5G throughput results after dropping constant metadata features.

Table 5.1: 5G Dataset Throughput Prediction Results (Target: `actual_throughput_mbps`)

Model	MAE	RMSE	R^2	CV R^2
DummyRegressor	36.90	43.88	-0.007	-0.011
Linear Regression	10.55	14.15	0.895	0.851
SVR (RBF)	5.27	8.16	0.965	0.925
Random Forest	2.76	6.13	0.980	0.936
XGBoost	3.20	6.32	0.979	0.934

The 5G dataset results show that supervised ML can predict achieved throughput well from radio and traffic configuration features. Random Forest gives the best held-out R^2 , while XGBoost is close. However, this dataset contains only one application type and no Gold/Silver/Bronze class variation. Therefore, it is useful as a link-capacity prediction experiment, but it cannot answer the per-class QoS allocation question on its own.

5.2 Layer 2: BNN-UPC Delay Prediction

5.2.1 Dataset and Setup

The BNN-UPC dataset was generated using BNNetSimulator. It contains 29,280 per-flow records. Each flow has a QoS class label: Gold, Silver, or Bronze. The target used for modelling is `log_avg_delay`, because raw delay is heavy-tailed.

The feature set includes offered bandwidth, ToS, QoS class, routing hops,

5.2 Layer 2: BNN-UPC Delay Prediction

topology size, traffic distribution, scheduling policy, and queue weights. Outcome columns such as raw delay, jitter, loss rate, delay percentiles, and actual bandwidth were removed from model inputs.

5.2.2 Global Model Results

Table 5.2 shows the global BNN-UPC results on `log_avg_delay`.

Table 5.2: BNN-UPC Global Results (Target: `log_avg_delay`)

Model	MAE	RMSE	R^2	CV R^2
Linear Regression	0.227	0.415	0.685	0.690
SVR (RBF)	0.174	0.391	0.721	0.721
Random Forest	0.181	0.364	0.758	0.764
XGBoost	0.185	0.359	0.765	0.766
PyTorch MLP	0.185	0.357	0.766	0.769

The PyTorch MLP is marginally ahead of XGBoost on the global score, but the difference is very small and within cross-validation noise: the five-fold cross-validated R^2 is 0.766 ± 0.008 for XGBoost and 0.769 ± 0.008 for the MLP (mean $\pm$ standard deviation across folds). The fold-level variability is much larger than the mean difference. This means the neural-network path is competitive, while XGBoost remains an important explainable tree-based baseline for this tabular dataset. The useful comparison is therefore not only which model has the highest global score, but also whether the MLP and XGBoost behave similarly for each QoS class. Section 5.3.4.1 tests this difference formally.

5.3 QoS-Aware Evaluation

A single global metric is not enough for BNN-UPC. Bronze delay is much more volatile than Gold and Silver delay. If all rows are pooled together, Bronze errors can dominate the score and hide how the model behaves for high-priority traffic.

Therefore, the XGBoost model was evaluated using out-of-fold predictions and sliced by QoS class, scheduling scenario, and scheduling policy. This provides the current explainable baseline for class-aware evaluation. The same class-aware evaluation was then applied to the PyTorch MLP so that the neural model is assessed per class, not only through a global score.

5.3.1 Per-Class Results

Table 5.3 shows per-class results. The metric values are on the delay scale in milliseconds, except R^2 , which is computed on the log-delay target.

Table 5.3: BNN-UPC Per-Class Evaluation using XGBoost

QoS Class	Rows	MAE Delay	RMSLE	R^2 Log Delay
Gold	4,592	3.05 ms	0.160	0.914
Silver	8,991	3.48 ms	0.168	0.908
Bronze	15,697	20.68 ms	0.451	0.704

The practical implication is significant. Gold and Silver, the classes that carry the most SLA-sensitive traffic, are predicted accurately ($R^2 = 0.914$ and 0.908), so the model is most reliable exactly where reliability matters most. Bronze, a best-effort class, is much harder ($R^2 = 0.704$) because its delay is dominated by transient queue contention rather than by the static configuration the model can observe (this mechanism is analysed in Chapter 6). Crucially, a single global score would have hidden this gap; only per-class evaluation exposes it, which is why

QoS-aware evaluation is essential for SD-WAN rather than an optional extra.

5.3.2 Policy and Scenario Results

Table 5.4 shows that the mixed-policy scenario is the hardest to predict.

Table 5.4: BNN-UPC Scenario-Specific Evaluation using XGBoost

Scenario	Rows	MAE Delay	R^2 Log Delay
A_wfq_fixed	6,560	6.16 ms	0.796
B_wfq_profiles	8,240	12.01 ms	0.758
C_mixed_policy	7,600	21.32 ms	0.743
D_mixed_equal	6,880	9.95 ms	0.779

Table 5.5 shows results sliced by scheduling policy. Strict Priority (SP) achieves the best R^2 because SP flows follow a deterministic priority order that the model can learn well. WFQ is harder because queue weights vary across profiles and interact with traffic load in more complex ways.

Table 5.5: BNN-UPC Scheduling Policy Evaluation using XGBoost (v1 Dataset). Policies: SP (Strict Priority), DRR (Deficit Round Robin), WFQ (Weighted Fair Queueing).

Policy	Rows	MAE (ms)	RMSLE	R^2
SP	8,120	8.27	0.282	0.806
DRR	1,840	8.01	0.302	0.678
WFQ	19,320	14.90	0.378	0.756

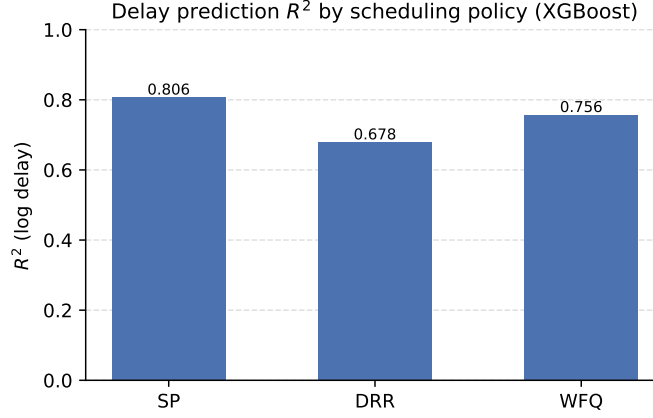


Figure 5.1: Delay prediction R^2 by scheduling policy (XGBoost). Strict Priority is the easiest to predict because its priority order is deterministic; DRR is the hardest.

5.3.3 SLA Violation Precision

For SD-WAN use, exact delay prediction is not the only concern. The model should also identify when a flow is likely to violate an SLA. The continuous delay prediction was converted into a binary trigger using these thresholds: Gold > 30 ms, Silver > 50 ms, and Bronze > 60 ms.

The Bronze threshold was tuned from an initial value of 100 ms to 60 ms following a sensitivity sweep. Table 5.6 summarises how precision, recall, and F1 change as the Bronze threshold is lowered. Lowering the threshold increases the number of flows treated as violations in the trigger evaluation, making the alert more sensitive and improving recall. The 60 ms value was chosen because it keeps Bronze strictly more lenient than Silver (50 ms) while significantly improving recall.

5.3 QoS-Aware Evaluation

Table 5.6: Bronze SLA Threshold Sensitivity (Gold=30 ms, Silver=50 ms fixed)

Bronze Threshold	Precision	Recall	F1
100 ms	0.726	0.368	0.488
70 ms	0.614	0.476	0.536
60 ms	0.619	0.568	0.592
50 ms	0.582	0.735	0.650

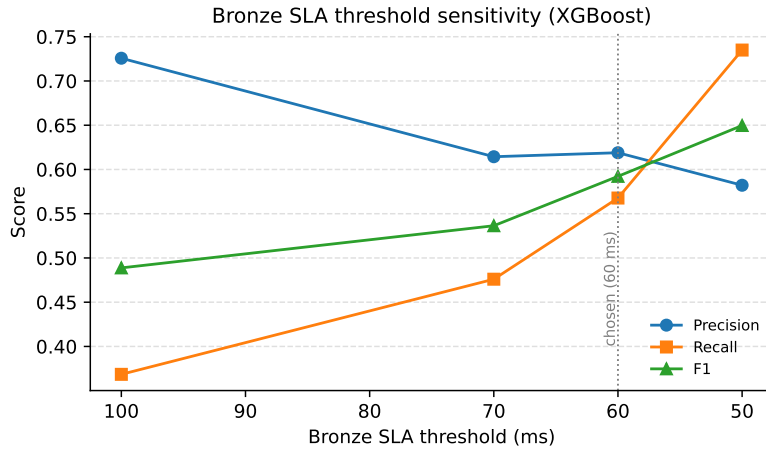


Figure 5.2: Bronze SLA threshold sensitivity for XGBoost. As the threshold is lowered from left to right, recall rises and precision falls. The chosen 60 ms value balances the two while staying more lenient than Silver (50 ms).

Table 5.7 shows the final SLA violation trigger results using the tuned thresholds.

Table 5.7: SLA Violation Trigger Results using XGBoost (Tuned Thresholds)

QoS Class	Threshold	Precision	Recall	F1
Gold	30 ms	0.994	0.899	0.944
Silver	50 ms	0.823	0.639	0.720
Bronze	60 ms	0.619	0.568	0.592

Gold precision remains very high (0.994), meaning the model is correct 99.4% of the time when it predicts a Gold SLA violation. This is critical for SD-WAN policy triggers, because false positives for Gold could cause unnecessary rerouting of high-priority traffic.

Silver sits between the two other classes: precision is good (0.823), but recall is only 0.639, meaning the model misses roughly one in three actual Silver SLA violations. This recall is not fully satisfactory. Silver carries medium-priority traffic, so a 36% miss rate is a genuine operational gap rather than a comfortable margin, and it is markedly weaker than Gold’s 0.899 recall. Unlike Bronze, however, the Silver threshold was never adjusted from its default of 50 ms; only the Bronze threshold went through the sensitivity sweep in Table 5.6. Whether a similar sweep can recover some of Silver’s missed recall without an unacceptable loss of precision remains untested. It is therefore added to Future Work as a natural extension of the threshold-tuning already carried out for Bronze.

Bronze recall improves from 0.368 (at 100 ms) to 0.568 (at 60 ms), a 55% improvement, while precision drops only modestly from 0.726 to 0.619. The threshold adjustment makes the alerting system more sensitive for Bronze, which is appropriate given that Bronze is best-effort traffic and missing a violation is more acceptable than for Gold.

5.3.4 Model Comparison: XGBoost vs MLP vs FT-Transformer

The results above use XGBoost as the strongest explainable tree-based baseline. To evaluate the neural-network direction of the project, the same QoS-aware analysis was repeated for two neural models of increasing sophistication:

- A **PyTorch feedforward MLP** (three hidden layers of 128, 64, and 32

units, ReLU activations, dropout 0.1, AdamW optimiser, early stopping).

- An **FT-Transformer** (Feature Tokenizer + Transformer), a modern attention-based architecture designed specifically for tabular data. Each feature (numeric or categorical) is converted into a learned token, a [CLS] token is prepended, three pre-norm Transformer blocks (eight attention heads, token dimension 64) apply self-attention across the feature tokens, and the final [CLS] representation is mapped to the predicted delay.

For all three models the outer out-of-fold evaluation protocol and metric functions are identical; the two neural models additionally use an internal validation split inside each training fold for early stopping. This makes the per-class numbers directly comparable while avoiding validation leakage. Including the FT-Transformer directly addresses the question of whether a modern deep-learning architecture, rather than a plain MLP, can outperform the tree-based baseline on this problem.

Table 5.8 compares per-class delay prediction.

5.3 QoS-Aware Evaluation

Table 5.8: Per-Class Delay Prediction: XGBoost vs MLP vs FT-Transformer (v1 Dataset, 5-fold out-of-fold)

QoS Class	Model	MAE (ms)	RMSLE	R^2
Gold	XGBoost	3.05	0.160	0.914
Gold	MLP	3.08	0.161	0.912
Gold	FT-Transformer	3.20	0.172	0.900
Silver	XGBoost	3.48	0.168	0.908
Silver	MLP	3.42	0.169	0.908
Silver	FT-Transformer	3.53	0.178	0.898
Bronze	XGBoost	20.68	0.451	0.704
Bronze	MLP	20.60	0.450	0.705
Bronze	FT-Transformer	20.52	0.452	0.703

On delay prediction all three models are practically very similar: per-class R^2 agrees to within about 0.01, and on the hardest class (Bronze) all three land at $R^2 \approx 0.70$. Neither neural model beats XGBoost; the FT-Transformer is in fact marginally behind on Gold and Silver (with reasonable but not exhaustive hyper-parameter tuning). This is consistent with the well-documented finding that gradient-boosted trees remain highly competitive with, and often superior to, deep-learning models on small, static, tabular feature sets such as this one.

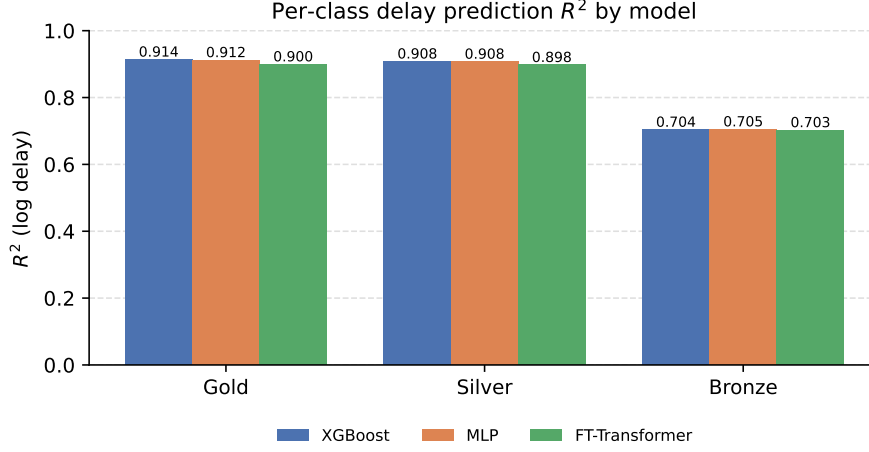


Figure 5.3: Per-class delay prediction R^2 for the three models. All three converge to almost identical performance per class, and all reach the same ceiling on the hard Bronze class ($R^2 \approx 0.70$), indicating that the limiting factor is the static feature set rather than model capacity.

The most important observation is that all three model families converge to the same per-class ceiling. This strongly suggests that the limiting factor is the *feature set*, not the choice of model: the static configuration features explain Gold and Silver delay very well but cannot fully capture Bronze’s burst-driven variance, regardless of model capacity.

Table 5.9 compares SLA violation detection at the tuned thresholds.

5.3 QoS-Aware Evaluation

Table 5.9: SLA Violation Detection: XGBoost vs MLP vs FT-Transformer (Tuned Thresholds)

QoS Class	Threshold	Model	Precision	Recall	F1
Gold	30 ms	XGBoost	0.994	0.899	0.944
Gold	30 ms	MLP	0.979	0.908	0.942
Gold	30 ms	FT-Transformer	0.980	0.898	0.937
Silver	50 ms	XGBoost	0.823	0.639	0.720
Silver	50 ms	MLP	0.791	0.715	0.751
Silver	50 ms	FT-Transformer	0.839	0.583	0.688
Bronze	60 ms	XGBoost	0.619	0.568	0.592
Bronze	60 ms	MLP	0.591	0.608	0.599
Bronze	60 ms	FT-Transformer	0.597	0.580	0.589

On SLA detection the three models again behave similarly, with only small differences in the precision–recall balance. The MLP leans slightly towards higher recall on the harder Silver and Bronze classes, while XGBoost and the FT-Transformer are marginally more precision-oriented. Across all three models Gold precision stays very high (0.98–0.99), which is the most important property for SD-WAN policy triggers because false positives on high-priority traffic would cause unnecessary rerouting.

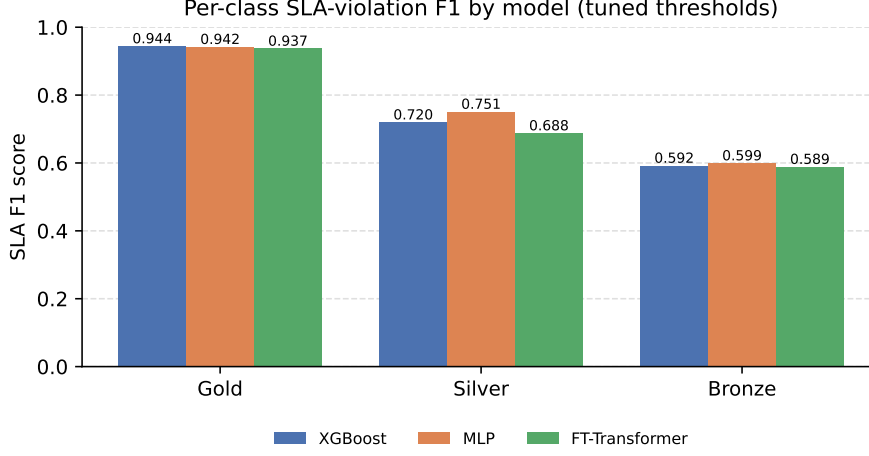


Figure 5.4: Per-class SLA-violation F1 score for the three models at the tuned thresholds (Gold 30 ms, Silver 50 ms, Bronze 60 ms). The models behave similarly; differences reflect small shifts in the precision–recall balance rather than a clear winner.

5.3.4.1 Statistical Significance of the XGBoost–MLP Difference

To test whether the small global gap between XGBoost and the MLP is real, both models were trained on the *same* five cross-validation folds and their per-fold R^2 values were compared with paired tests. This fold-level paired comparison is the standard way to compare two learners under cross-validation; a per-row test was deliberately avoided because, with roughly 29,000 rows, even a negligible difference would register as significant.

The MLP was higher by a mean of only 0.0006 R^2 (XGBoost 0.765, MLP 0.766 averaged over the five folds). A paired t -test returned $p = 0.620$, while the rank-based Wilcoxon signed-rank test returned $p = 0.8125$. Both tests are non-significant at the 0.05 level, and the difference (0.0006) is far smaller than the fold-to-fold standard deviation (≈ 0.016). The practical conclusion is that the two models are equivalent for this task: any difference is an order of magnitude

5.4 Multi-Metric QoS Evaluation (v1 Dataset)

smaller than normal resampling variation, and with only five folds the tests have limited power. This supports, with evidence rather than by inspection, the claim that the neural model does not meaningfully outperform the tree-based baseline.

Overall, XGBoost is retained as the recommended model: it matches or slightly exceeds both neural models on accuracy while offering interpretability (feature importance) and substantially lower training cost (seconds versus minutes per fold for the FT-Transformer). The progression from a tree-based baseline through a feedforward MLP to a modern tabular Transformer confirms that the strong per-class results are a property of the data and features, not an artefact of a single model family, and that added model complexity does not improve QoS delay prediction on this dataset.

5.4 Multi-Metric QoS Evaluation (v1 Dataset)

Beyond average delay, the BNN-UPC v1 dataset was used to evaluate delay_p90 and jitter per QoS class. The same out-of-fold XGBoost pipeline was used for both targets, with a log transform applied before training.

5.4.1 delay_p90 Results

Table 5.10 shows the per-class results for 90th-percentile delay prediction.

Table 5.10: Per-Class delay_p90 Prediction Results using XGBoost (v1 Dataset)

QoS Class	Rows	MAE (ms)	RMSLE	R^2
Gold	4,592	6.64 ms	0.205	0.635
Silver	8,991	7.29 ms	0.215	0.657
Bronze	15,697	38.22 ms	0.499	0.258

Gold and Silver show moderate predictability ($R^2 \approx 0.64$ – 0.66), which is lower

than for average delay. This is expected because the 90th percentile captures rarer, more variable congestion events. Bronze tail delay is much harder to predict ($R^2 = 0.258$), following the same pattern as average delay but amplified.

5.4.2 Jitter Results

Table 5.11 shows the per-class results for jitter prediction.

Table 5.11: Per-Class Jitter Prediction Results using XGBoost (v1 Dataset)

QoS Class	Rows	MAE (ms)	RMSLE	R^2
Gold	4,592	0.14 ms	0.176	0.119
Silver	8,991	0.16 ms	0.190	0.084
Bronze	15,697	3.11 ms	0.678	0.014

Jitter R^2 is very low across all classes (0.01–0.12). This is a meaningful finding rather than a modelling failure. The BNN-UPC features describe static network configuration: queue weights, scheduling policy, and offered bandwidth. These features determine mean delay well but do not capture the moment-to-moment traffic burstiness that drives jitter. Jitter prediction would require dynamic traffic features such as instantaneous queue occupancy or inter-arrival time statistics, which are not present in this dataset. This is discussed further as a limitation and future work direction.

5.5 Bronze Packet Loss Analysis

In the v1 dataset, packet loss is realistic: Gold and Silver flows experience near-zero loss (4 and 93 loss events respectively out of 13,583 combined rows), while Bronze flows experience non-zero loss in 1,684 out of 15,697 rows (10.7%). This

matches the expected behaviour of a network where high-priority traffic (Gold, Silver) is protected and only best-effort Bronze traffic suffers under load.

Following supervisor guidance, packet loss evaluation is restricted to Bronze traffic only. Gold and Silver loss events are too rare and too small to train a reliable model on, and artificially forcing loss onto high-priority classes would not reflect a realistic network scenario.

5.5.1 Regression Attempt

Regression on `packet_loss_rate` for Bronze flows produced $R^2 = -0.130$. This is a meaningful finding: packet loss events in this dataset are driven by rare, bursty buffer overflow events. The static network configuration features (queue weights, scheduling policy, offered bandwidth) determine mean delay well but cannot capture when a particular burst will exceed the buffer. Predicting the exact loss rate requires dynamic features such as instantaneous buffer occupancy or inter-arrival time statistics, which are not present in the dataset.

5.5.2 Binary Loss Classification for Bronze

A more appropriate framing is binary classification: predicting whether a Bronze flow will experience any packet loss at all ($\text{loss} > 0$). This converts the problem from estimating an unpredictable rate to a detection task. The class balance is 89.3% no-loss versus 10.7% loss, so class weighting (`scale_pos_weight = 8.3`) is applied.

Table 5.12 shows 5-fold stratified cross-validation results for XGBoost and Random Forest.

5.5 Bronze Packet Loss Analysis

Table 5.12: Bronze Binary Loss Classification Results (v1 Dataset, 5-fold CV)

Model	Precision	Recall	F1	AUC-ROC
XGBoost (default threshold)	0.777	0.328	0.461	0.914
XGBoost (class-weighted)	0.342	0.865	0.490	0.916
Random Forest (balanced)	0.663	0.389	0.490	0.895

The AUC-ROC of 0.916 for class-weighted XGBoost confirms the model has strong discriminative ability: it can rank Bronze flows by loss probability effectively. The low recall at the default 0.5 decision threshold (0.328) is a threshold choice, not a model failure. Using class weighting raises recall to 0.865, but this comes at a real operational cost, not just a lower precision number (0.342): roughly two out of every three flagged flows do not actually experience loss. A false-alert share this high risks alert fatigue, where operators or downstream automation begin to distrust or deprioritise the system’s warnings, which would undermine the alerting system more than the missed-detection problem it was meant to solve. Neither the default-model operating point nor the class-weighted operating point is unambiguously correct; the right operating point depends on the true operational cost of a missed loss event versus a false alarm, and should be chosen deliberately, for example via a precision-recall curve analysis rather than a default 0.5 cut-off (Future Work).

This shows that while exact loss-rate regression is not feasible from static features, the model can reliably identify which Bronze flows are at elevated risk of experiencing packet loss.

5.6 Layer 3: Allocation Recommender

The final experiment uses the trained BNN-UPC delay model as a surrogate model for allocation decisions. The recommender is designed so that it can use either an explainable model such as XGBoost or a neural model such as the PyTorch MLP. The current reported allocation results use the trained delay model already evaluated in this chapter. Several WFQ profiles were tested: 60/30/10, 50/40/10, 33/33/34, 25/65/10, and 80/15/5. Each profile was applied to the traffic pattern, and the model predicted the expected delay for Gold, Silver, and Bronze.

The default SLA thresholds for this allocation experiment were Gold < 20 ms, Silver < 50 ms, and Bronze < 100 ms. This recommender run keeps the original allocation thresholds; rerunning the recommender with the tuned Bronze threshold of 60 ms is left as a follow-up comparison.

Table 5.13: Layer 3 WFQ Allocation Recommendation Results

Rank	Profile	Feasible	Gold Mean	Silver Mean	Bronze Mean
1	60/30/10	No	24.47 ms	24.22 ms	30.81 ms
2	80/15/5	No	24.99 ms	30.62 ms	31.27 ms
3	25/65/10	No	25.02 ms	24.28 ms	30.81 ms
4	33/33/34	No	25.12 ms	24.55 ms	33.92 ms
5	50/40/10	No	25.27 ms	24.80 ms	30.81 ms

No tested profile fully satisfies the strict Gold threshold of 20 ms. The recommender therefore selects 60/30/10 as the least-violating profile. This is still useful because the system can show both a recommendation and an infeasibility warning. In practice, this could tell an SD-WAN controller that changing WFQ weights alone may not be enough and that rerouting or extra capacity

may be needed. A future comparison should run the same profile search using the PyTorch MLP and check whether the neural model recommends the same allocation.

5.6.1 Towards a Deployment Architecture

The recommender in this thesis runs offline on a fixed dataset, but its design maps onto a realistic closed-loop control system for a production SD-WAN. Figure 5.5 sketches how the components would fit together. The SD-WAN controller continuously collects telemetry (per-flow delay, loss, and offered load) from the edge routers. This telemetry is fed to the trained QoS model, which predicts per-class delay under the current conditions. The recommender then evaluates candidate WFQ profiles using the model as a surrogate and returns either a recommended profile or an infeasibility signal. Finally, the controller enacts the decision: it pushes the new QoS policy to the edge routers when a feasible profile exists, or escalates to a different control action (rerouting or adding capacity) when none does. The loop then repeats as new telemetry arrives.

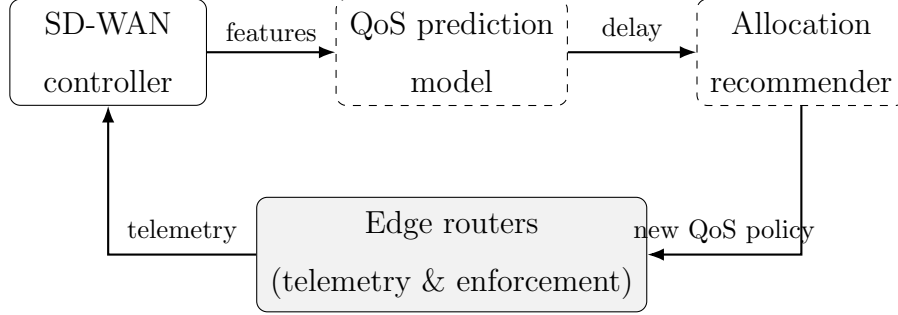


Figure 5.5: A closed-loop deployment architecture for the recommender. Edge-router telemetry feeds the controller, which drives the prediction model and recommender; the resulting policy (or an escalation when no profile is feasible) is pushed back to the edge routers, and the loop repeats. The dashed offline components of this thesis correspond to the “QoS prediction model” and “allocation recommender” blocks.

Two properties of the design are worth noting. First, the model must be cheap enough to run inside this loop, which reinforces the choice of XGBoost: it infers in milliseconds and retrains in seconds, unlike the neural models. Second, the infeasibility signal is what makes the loop safe: rather than silently applying a profile that cannot meet the SLA, the system tells the controller to escalate, so that the automation fails loudly rather than degrading service quietly.

5.7 Summary of Findings

The completed experiments show seven main findings:

- The 5G dataset is useful for link capacity prediction, but not for per-class QoS allocation.
- BNN-UPC provides a useful per-class QoS scheduling dataset. Global metrics hide class-level differences; Bronze is much harder to predict than Gold

and Silver.

- QoS-aware evaluation with per-class MAE, RMSLE, and SLA violation precision is essential for reporting honest model behaviour for SD-WAN.
- `delay_p90` is moderately predictable ($R^2 \approx 0.64$ – 0.66 for Gold/Silver), but jitter is not predictable from static configuration features ($R^2 < 0.12$). This is a feature gap, not a modelling failure.
- Bronze SLA recall improves significantly when the violation threshold is tuned from 100 ms to 60 ms (Recall: $0.368 \rightarrow 0.568$, F1: $0.489 \rightarrow 0.592$). The 60 ms threshold was selected to keep Bronze strictly more lenient than Silver (50 ms) while providing a meaningful recall improvement.
- Packet loss is only meaningful for Bronze traffic in a realistic network. Regression on `packet_loss.rate` is not feasible from static features ($R^2 = -0.130$), but binary classification (loss vs. no loss) achieves AUC-ROC = 0.916 for Bronze flows, confirming the model can identify loss-prone flows reliably.
- Three model families were compared per class under an identical evaluation protocol: XGBoost, a feedforward MLP, and a modern attention-based FT-Transformer. All three converge to the same per-class ceiling (Bronze $R^2 \approx 0.70$ for every model), and neither neural model beats XGBoost. This shows the limiting factor is the static feature set, not model capacity, and XGBoost is retained as the recommended model for its equal accuracy, interpretability, and far lower training cost.

Chapter 6

Discussion

The previous chapter reported what the models achieved. This chapter explains why those results occurred, which is where the practical and scientific value of the work lies. Five questions are addressed: why some QoS classes are far easier to predict than others, why deep learning did not outperform a tree-based model, why jitter and packet loss resist prediction, why link throughput is easier to predict than per-class delay, and what all of this means for a real SD-WAN deployment.

6.1 Why Per-Class Predictability Differs

Gold and Silver delay are predicted well ($R^2 \approx 0.91$), while Bronze is much harder ($R^2 \approx 0.70$). The reason lies in how the scheduler treats each class. Gold and Silver traffic is protected by high queue priority, so its delay is dominated by the deterministic part of the system: the number of hops on the path and the configured queue weights. The SHAP analysis in Figure 4.2 confirms this, showing that `routing_hops` is by far the strongest predictor, followed by network size and traffic class. These are exactly the static, configuration-level quantities

6.2 Why Deep Learning Did Not Outperform Tree-Based Models

the model has access to, so it can learn them well.

Bronze traffic is best-effort. It is served only after the higher-priority classes, so its delay is governed by transient queue contention: short bursts of higher-priority traffic that momentarily starve the Bronze queue. This burstiness is a dynamic, moment-to-moment property that the static feature set does not describe. The model can predict the average behaviour of Bronze traffic but not the variance introduced by contention, which is why its error is both larger and more heavy-tailed. The same mechanism explains why the Bronze tail delay (`delay_p90`) is harder still: the tail is precisely where contention events dominate.

6.2 Why Deep Learning Did Not Outperform Tree-Based Models

A central result is that the MLP and the FT-Transformer matched, but did not beat, XGBoost, and that the three models converged to the same per-class ceiling. This is consistent with a growing body of evidence that gradient-boosted decision trees remain highly competitive on small, static, tabular datasets. Grinsztajn et al. [4] and Shwartz-Ziv and Armon [9] both show that tree ensembles typically equal or exceed deep networks on tabular problems, because trees handle irregular, non-smooth feature interactions and are far less sensitive to uninformative features and to the absence of large training sets.

The present dataset has exactly these characteristics: around 29,000 rows, eleven mostly static features, and no temporal or relational structure for an attention mechanism to exploit. The FT-Transformer’s strength, learning interactions across many feature tokens, brings little benefit when the target is already well explained by a few dominant features (chiefly path length). The convergence of all three model families is therefore not a failure of the neural models; it is

evidence that the ceiling is set by the information available in the features rather than by model capacity. This reframes the negative result as a meaningful finding about the problem, not the method.

6.3 Why Jitter and Packet Loss Resist Prediction

Jitter is almost unpredictable from the available features ($R^2 < 0.12$), and exact packet-loss rate cannot be regressed at all ($R^2 = -0.130$ for Bronze). Both outcomes have the same root cause as the Bronze delay difficulty, in a stronger form. Jitter is the variation of delay, and packet loss is caused by momentary buffer overflow; both are driven almost entirely by short-timescale traffic burstiness rather than by the average configuration. The static feature set records what the network is configured to do, not what the traffic is doing at any instant, so the information needed to predict these quantities is simply not present. This is a limitation of feature observability, and it is why the packet-loss problem was reframed as binary classification (will a flow experience any loss?), which succeeds with an AUC-ROC of 0.916: identifying which flows are at risk needs only the average conditions, whereas quantifying the exact loss needs the missing dynamic information.

6.4 Why Link Throughput Is Easier to Predict Than Per-Class Delay

The Layer 1 throughput task reaches a much higher accuracy (held-out $R^2 \approx 0.98$) than the Layer 2 delay task. This is expected. Achieved throughput on a radio

link is largely a direct, monotonic function of the measured channel and traffic conditions, so the input features carry a strong and stable signal about the output. Per-class delay, by contrast, is an emergent property of queueing interactions between competing traffic classes at each hop, which is a more indirect and higher-variance relationship. The contrast is itself informative: it shows that the difficulty is not machine learning in networking per se, but the specific gap between static configuration features and the dynamic phenomena that drive per-class QoS.

6.5 Operational Implications for SD-WAN

Several findings matter for a real deployment. First, the very high Gold SLA-violation precision (0.994) is the most operationally useful number in the thesis: an SD-WAN controller can act on a predicted Gold violation with high confidence that it is real, avoiding the unnecessary rerouting that a false alarm on high-priority traffic would cause. Second, because XGBoost matches the neural models while remaining interpretable and training in seconds rather than minutes, it is the sensible choice for a system that must retrain and infer close to real time. Third, the allocation recommender’s ability to report that no weight profile can satisfy the SLA is arguably more valuable than the recommendation itself: it tells the controller when the problem cannot be solved by tuning queue weights and that rerouting or additional capacity is required. Taken together, these results support a practical division of labour in which learned models handle the well-conditioned, high-priority predictions and flag the cases where a different control action is needed.

Chapter 7

Limitations

This chapter states the main limitations of the work explicitly, so that the results are not over-interpreted.

7.1 Data Availability and Realism

No real-world enterprise SD-WAN dataset with labelled QoS metrics was available for this project, so the work relied on public and simulator-generated data. The 5G dataset provides real measurements but contains no Gold/Silver/Bronze QoS-class structure. The BNN-UPC dataset does provide per-class behaviour, but it is produced by a network simulator (BNNetSimulator) rather than measured on a live network. The official BNN-UPC download links were unavailable, so the dataset was generated locally using the official simulator and the same output format. CICIDS2017 was discarded because it is an intrusion-detection dataset rather than a QoS dataset, and MAWI was not used because it lacks direct QoS-class labels. The findings should therefore be read as a reproducible proof of concept, not as a claim of production-grade generalisation.

7.2 Static Features and the Bronze Ceiling

All models were trained on static configuration features (offered load, routing hops, scheduling policy, queue weights, and ToS class). These features explain Gold and Silver delay well, but they cannot capture the short-term traffic burstiness that drives Bronze delay variance, tail delay, jitter, and packet loss. This limit was confirmed empirically: three different model families, and additional engineered interaction features, all converged to the same per-class ceiling (Bronze $R^2 \approx 0.70$). The limitation is therefore one of feature observability rather than model capacity.

7.3 Offline Evaluation

The evaluation is entirely offline. The metrics used (per-class error, RMSLE, SLA-violation precision and recall, and the allocation recommender) are closer to operational SD-WAN use than a single global score, but they do not capture live behaviour such as failover, multi-path routing, or the feedback effects of acting on a prediction. Validation inside a live SD-WAN controller remains future work.

7.4 Computational Constraints

Model training and tuning were carried out on a single laptop without dedicated GPU acceleration. This limited the extent of neural-network hyper-parameter search and the size of the simulated dataset. Because the tree-based and neural models reached the same per-class ceiling, this constraint is unlikely to have changed the main conclusions, but a larger compute budget would allow a wider search and more simulated data.

Chapter 8

Conclusion

This thesis investigated whether machine learning can predict QoS in SD-WAN-like environments and, going a step beyond prediction, whether those predictions can support a concrete bandwidth-allocation decision. The work was organised as three connected layers: link-capacity prediction, per-class QoS prediction, and allocation recommendation. The three research questions can now be answered directly.

RQ1 (can supervised ML accurately predict per-class QoS?). Yes, but conditionally. Gold and Silver average delay are predicted accurately ($R^2 = 0.914$ and 0.908), whereas Bronze delay, tail delay, jitter, and exact packet loss are not, because they are driven by short-timescale traffic burstiness that the static configuration features do not capture. The value of the answer lies in its precision: the per-class, multi-metric evaluation is what makes this boundary between predictable and unpredictable QoS visible, where a single global score would have hidden it.

RQ2 (which model family offers the best balance?). For this small, static, tabular dataset, the tree-based XGBoost is the best choice. A feedforward MLP and a modern FT-Transformer matched it but did not beat it, with differ-

ences within cross-validation noise and confirmed non-significant by a paired test, while XGBoost is more interpretable and far cheaper to train and run. This is a useful negative result: added model complexity did not help, because the ceiling is set by the features, not the model.

RQ3 (can predicted QoS support an allocation decision?). Yes. The Layer 3 recommender uses the delay model as a surrogate to rank WFQ bandwidth profiles, and when no profile can satisfy the SLA it reports infeasibility rather than a false recommendation. That honest signal, telling the controller when tuning weights cannot solve the problem, is itself an actionable decision-support output.

The literature review identified a broad range of AI and ML techniques applied to QoS prediction and SD-WAN traffic engineering, ranging from classical supervised learning to graph neural networks and reinforcement learning. The review confirmed that while deep learning and transformer-based models represent the current state of the art, interpretable baseline models remain essential as comparison benchmarks and are computationally feasible for near-real-time deployment.

The first dataset is the 5G dataset [7]. It is useful for predicting link capacity from radio and traffic features. On this dataset, tree-based and kernel models perform strongly, with Random Forest and XGBoost reaching a held-out test R^2 of about 0.98 (cross-validated $R^2 \approx 0.94$). This shows that the measured link conditions contain a strong signal for throughput prediction.

The second dataset is the BNN-UPC dataset generated using BNNetSimulator. This dataset is more important for the original research question because it includes Gold, Silver, and Bronze traffic classes under different scheduling policies and queue weights. On log-transformed average delay all evaluated models reach a similar global score (cross-validated $R^2 \approx 0.77$). A per-class comparison

of three model families, XGBoost, a feedforward MLP, and an FT-Transformer, showed that they converge to almost identical per-class performance, and neither neural model beat XGBoost. XGBoost is therefore retained as the recommended model because it matches the neural models while remaining interpretable and far cheaper to train.

The QoS-aware evaluation is a key part of the work. It shows that a single global score is not enough. Gold and Silver delay are predicted with low error, while Bronze is more difficult because its delay is more volatile and heavy-tailed. SLA violation precision is especially strong for Gold traffic, which is useful because Gold flows are normally the most important class for SD-WAN policy decisions.

The Layer 3 allocation recommender uses the trained BNN-UPC delay model to compare candidate WFQ weight profiles. In the tested traffic pattern, none of the profiles fully satisfy the strict Gold SLA threshold of 20 ms. The recommender therefore selects the least-violating profile and reports that weight adjustment alone may not be enough. This connects the predictive model back to the original SD-WAN question of how much bandwidth should be allocated to each QoS class, and it is where the thesis moves beyond prediction into decision support.

Taken together, the contribution of this work is not a single accurate model but a method: a per-class, QoS-aware way of evaluating QoS prediction, an honest comparison showing where deep learning does and does not help, and a recommender that turns predictions into a bounded control decision. This is a more useful and more defensible outcome for SD-WAN than a headline accuracy figure alone.

8.1 Real-World Impact or Benefit

The core value of this project lies in supporting proactive QoS management in SD-WAN environments through AI-driven predictions. In current networks, QoS violations are often addressed reactively, after user experience is already affected. This work is a step in that direction: it predicts per-class QoS behaviour from network configuration features offline, as a proof of concept that could later be driven by live telemetry.

Potential benefits, if integrated into a live system:

- **Proactive Network Management:** Helps operators act before congestion or SLA violations occur.
- **Cost Optimization:** Enables more informed bandwidth allocation and may reduce unnecessary overprovisioning.
- **Improved QoE (Quality of Experience):** Real-time predictions allow better support for critical applications like video calls, VoIP, and live collaboration tools.
- **Scalability:** The approach is suitable for large enterprise SD-WANs that span multiple locations and require dynamic QoS adjustments.

8.2 Future Work

Based on the current findings, future research should explore the following directions:

- **Bronze Loss Classifier Deployment:** The current Bronze binary loss classifier achieves $\text{AUC-ROC} = 0.916$. The next step is to select the optimal decision threshold (e.g. using precision-recall curve analysis) and integrate

the classifier alongside the delay model so that the SD-WAN controller receives both a delay prediction and a loss-risk score for each Bronze flow.

- **Neural Allocation Recommender:** Extend the Layer 3 recommender to use the PyTorch MLP, then compare per-class allocation results against XGBoost using Gold/Silver/Bronze SLA thresholds.
- **Better Bronze Delay Prediction:** Bronze delay is harder to predict than Gold and Silver ($R^2 = 0.704$ vs. 0.914 and 0.908). A dedicated model with class-weighted training or a separate Bronze-only model may reduce the Bronze MAE further. The tuned SLA threshold (60 ms) already improves recall, but better per-class models remain worth exploring.
- **Silver SLA Threshold Tuning:** Silver SLA recall is only 0.639, and unlike Bronze its 50 ms threshold has not been tuned. Applying the same threshold-sensitivity sweep used for Bronze (Table 5.6) to Silver would test whether some of its high precision (0.823) can be traded for improved recall.
- **Jitter Prediction with Dynamic Features:** Jitter is not predictable from static configuration features. Future work should explore adding dynamic traffic features such as instantaneous queue depth or inter-arrival time statistics.
- **More Allocation Profiles:** Test a wider search space of WFQ weights and include rerouting or extra-capacity decisions when no feasible weight profile exists.
- **Comparison Against a Standard Controller Policy:** The Layer 3 recommender currently ranks candidate WFQ profiles against each other. A direct comparison against the allocation a standard, rule-based SD-WAN controller would choose (for example a fixed default profile) would quantify

the improvement the learned recommender provides, addressing a gap noted in the literature review.

- **Dynamic and Online Learning:** Move from static, batch-trained models to online learning that updates as live telemetry arrives, so the predictor tracks changing traffic instead of being retrained from scratch.
- **Reinforcement Learning for Control:** Combine the QoS predictor with reinforcement learning to close the loop, letting a policy agent learn allocation and rerouting actions directly from predicted QoS, subject to SLA constraints.
- **Real-Time Integration:** Integrate the predictive model into an SD-WAN controller or policy engine through vendor APIs (for example Cisco Catalyst Center/vManage or FortiManager), and validate it against a digital-twin or intent-based-networking environment before live deployment.
- **Cross-Domain Generalisation:** Test the approach on more topologies, real traffic traces, and additional public datasets such as MAWI, to measure how far results transport beyond the simulator.

By addressing these improvements, the project could evolve into a plug-in or microservice within SD-WAN platforms that provides predictive QoS as a built-in capability.

8.3 Final Remarks

AI and ML hold significant promise for improving QoS in SD-WAN by enabling proactive, predictive management rather than reactive troubleshooting. The contribution of this thesis is to push that promise one step past prediction: it shows

that the right question is not only how accurately QoS can be predicted, but for which classes and metrics prediction is reliable, which model earns its complexity, and how a prediction becomes a safe control decision. By evaluating QoS per class, comparing tree-based and deep-learning models honestly, and building a recommender that knows when it cannot help, this work advances AI-assisted SD-WAN QoS management from a modelling exercise towards trustworthy, deployable decision support. With further work on live telemetry, online learning, and controller integration, AI-driven QoS management can become a core, dependable feature of future network systems.

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